

Name:.....

Student
Number

Teacher:.....



Pymble Ladies' College

Mathematics Extension 1

HSC Trial Examination

Term 3 2023

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks 70

SECTION 1 – 10 marks (pages 1-4)

- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Answer each question on the multiple-choice answer sheet provided in the answer booklet.

SECTION II – 60 marks (pages 5-10)

- Attempt Questions 11-14
- Allow about 1 hours and 45 minutes for this section
- Answer each question in the appropriate space in the Answer Booklet. Extra writing pages are included at the end of each question.

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SECTION I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. Given the points $A(-2, -5)$ and $B(0, 7)$, which of the following represents the displacement vector $\overrightarrow{BA}$?
 - (A) $-2\mathbf{i} - 12\mathbf{j}$
 - (B) $2\mathbf{i} + 12\mathbf{j}$
 - (C) $-2\mathbf{i} + 2\mathbf{j}$
 - (D) $2\mathbf{i} + 2\mathbf{j}$

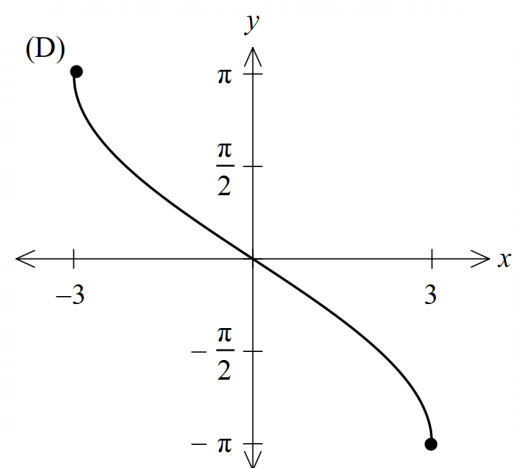
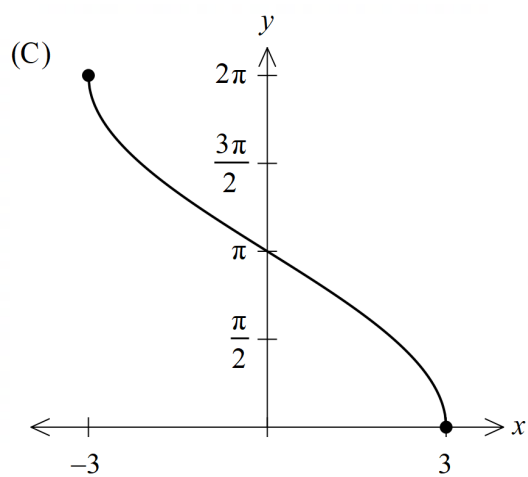
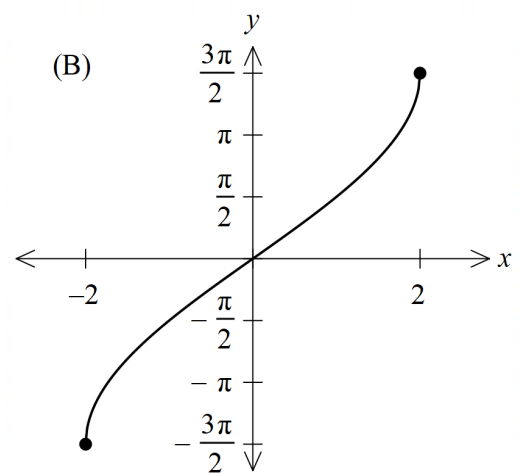
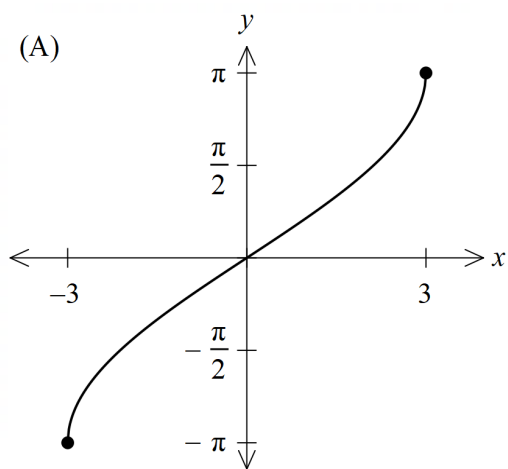
2. What is the anti-derivative of $\cos^2 2x$?
 - (A) $\frac{\cos^3 2x}{3} + C$
 - (B) $\frac{1}{2} \sin^2 2x + C$
 - (C) $\frac{x}{2} + \frac{\sin 4x}{8} + C$
 - (D) $\frac{1}{2} + \frac{1}{2} \cos 4x + C$

3. Which of the following expressions is the correct rate of change of the area of a circle (A) with respect to time (t)?
 - (A) $\frac{dr}{dt} = 2\pi r \times \frac{dA}{dt}$
 - (B) $\frac{dr}{dt} = 2\pi r \times \frac{dA}{dr}$
 - (C) $\frac{dA}{dr} = 2\pi r \times \frac{dt}{dr}$
 - (D) $\frac{dA}{dt} = 2\pi r \times \frac{dr}{dt}$

4. Which polynomial has a root at $x = 1$ of multiplicity 3?

- (A) $x^3 - 3x + 2$
- (B) $x^4 - 2x^3 + 2x - 1$
- (C) $x^3 + 4x^2 + 5x + 2$
- (D) $x^4 + 2x^3 - 2x - 1$

5. Which graph best represents $y = 3 \sin^{-1} \frac{x}{2}$?



6. Which of the following is the derivative of $y = \tan^{-1}[2f(x)]$?

(A) $\frac{dy}{dx} = \frac{1}{1+[f(x)]^2}$

(B) $\frac{dy}{dx} = \frac{2}{1+4[f(x)]^2}$

(C) $\frac{dy}{dx} = \frac{f'(x)}{1+4[f(x)]^2}$

(D) $\frac{dy}{dx} = \frac{2f'(x)}{1+4[f(x)]^2}$

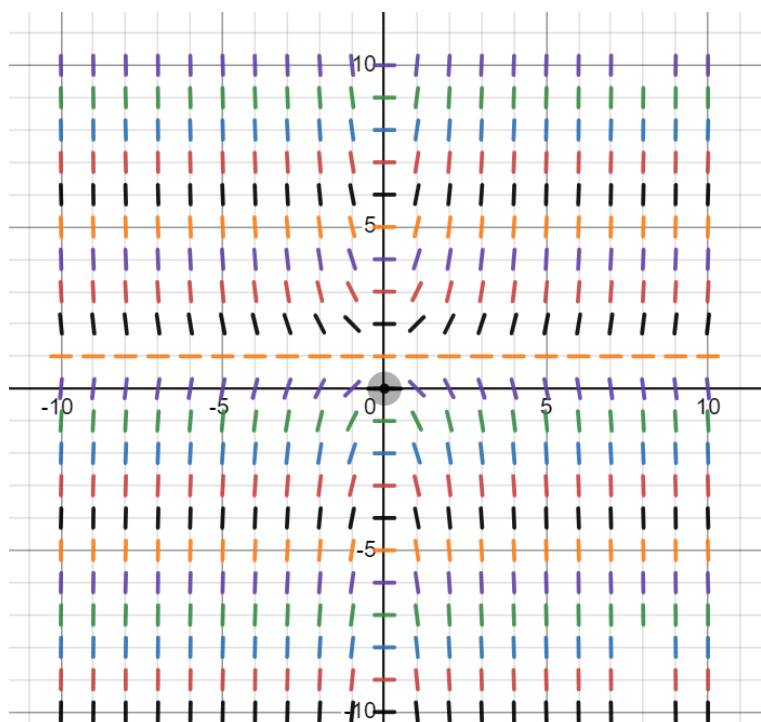
7. The direction field for a certain differential equation is shown below. Which of the following is the differential equation?

(A) $\frac{dy}{dx} = x^2 + y^2$

(B) $\frac{dy}{dx} = x^2 - y^2$

(C) $\frac{dy}{dx} = xy - x$

(D) $\frac{dy}{dx} = y^2 - x^2$



8. What is the value of $\sin^{-1}(\cos \alpha)$ where $\frac{3\pi}{2} < \alpha < 2\pi$?

(A) $2\pi - \alpha$

(B) $\alpha - 2\pi$

(C) $\frac{3\pi}{2} + \alpha$

(D) $\alpha - \frac{3\pi}{2}$

9. The equation $y = e^{ax}$ satisfies the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0.$$

What are the possible values of a ?

(A) $a = -2$ or $a = 3$

(B) $a = -1$ or $a = 6$

(C) $a = -3$ or $a = 2$

(D) $a = -6$ or $a = 1$

10. Students at a school are required to study 4 subjects from a selection of 10. The ten subjects are placed in two lines, each containing 5 subjects.

How many possible combinations of subjects exist if a student must take at least 1 subject from each line?

(A) 200

(B) 250

(C) 309

(D) 356

SECTION II

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section.

Answer these questions in the Answer Book provided.

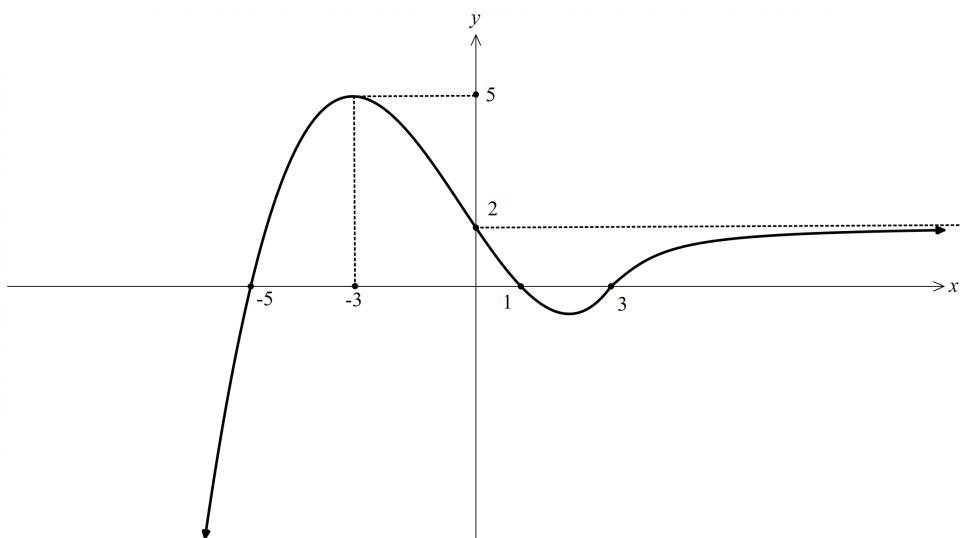
Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (17 marks)	Marks
(a) The polynomial $P(x) = 2x^4 - 11x^3 + 19x^2 - 13x + 3$ has roots α, β, γ and δ . Find the value of $\frac{1}{\alpha\beta\gamma} + \frac{1}{\alpha\beta\delta} + \frac{1}{\beta\gamma\delta} + \frac{1}{\alpha\gamma\delta}$	2
(b) Find $\int \frac{1}{4x^2 + 9} dx$.	2
(c) A square metal sheet is heated so that its side length is increasing at a uniform rate of 0.05 cms^{-1} . Find the rate its area is changing at the instant when the side length is 3.1 cm.	2
(d) Find $\int_0^{\frac{\pi}{6}} \sin 4x \cos 2x dx$.	3
(e) Given $\underline{a} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} -6 \\ 4 \end{bmatrix}$, find: (i) $ \underline{a} $ (ii) $\underline{a} \cdot \underline{b}$ (iii) the angle between $\underline{a}$ and $\underline{b}$, to the nearest degree. (iv) $\text{proj}_{\underline{b}} \underline{a}$	1 1 1 2

Question 11 continues on page 6

Question 11 (continued)

- (f) The diagram shows the graph of $y = f(x)$ which has a horizontal asymptote at $y = 2$.



In your answer booklet draw clear sketches of the following.
Show all relevant information.

(i) $y = |f(x)|$

1

(ii) $y = \frac{1}{f(x)}$

2

End of Question 11

Question 12 (17 marks)**Marks**

- (a) Find $\int \frac{3x}{\sqrt{9-x^2}} dx$ using the substitution $u = 9 - x^2$. 2
- (b) By expressing $\sqrt{3} \cos x - \sin x$ in the form $A \cos(x + \alpha)$, where $A > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$, 3
solve $\sqrt{3} \cos x - \sin x + 1 = 0$, for $0 \leq x \leq 2\pi$.
- (c) Paddington Bear hits a golf ball with velocity V m/s at an angle of projection θ degrees to the horizontal.
The golf ball's position at any time t , is given by the position vector
$$\underline{r}(t) = (Vt \cos \theta) \underline{i} + \left(Vt \sin \theta - \frac{1}{2}gt^2 \right) \underline{j}. \quad [\text{Do not prove this}]$$
- (i) Show that Paddington's golf ball reaches a maximum height of 2
$$\frac{V^2 \sin^2 \theta}{2g} \text{ metres.}$$
- (ii) What is the range of flight of the golf ball? 2
- (d) Prove by mathematical induction that $4^n + 14$ is divisible by 6 for all positive integers $n \geq 1$. 3
- (e) The polynomial $P(x) = x^3 + bx^2 + cx + d$ has roots 0, 4 and -4 .
- (i) Find b , c and d . 2
- (ii) Without calculus, sketch the graph of $y = P(x)$. 1
- (iii) Hence, or otherwise, solve the inequality $\frac{x^2 - 16}{x} \leq 0$. 2

End of Question 12

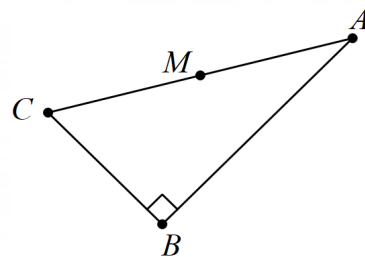
Question 13 (14 marks)**Marks**

- (a) What is the value of the term independent of x in the expansion of

2

$$\left(2 - \frac{3}{x^3}\right) \left(2x + \frac{5}{x^2}\right)^6 ?$$

- (b) ABC is a right-angled triangle. M is the midpoint of the hypotenuse AC , as shown below. Let $\overrightarrow{AM} = \underline{a}$ and $\overrightarrow{BM} = \underline{b}$.



NOT TO SCALE

- (i) Express $\overrightarrow{AB}$ and $\overrightarrow{BC}$ in terms of $\underline{a}$ and $\underline{b}$.
- (ii) Prove that M is equidistant from the three vertices of $\triangle ABC$.
- (c) A relation is defined by the parametric equations

$$x = 1 - \sin t, \quad y = \cos t \sin 2t, \quad \text{for } 0 \leq t \leq \pi.$$

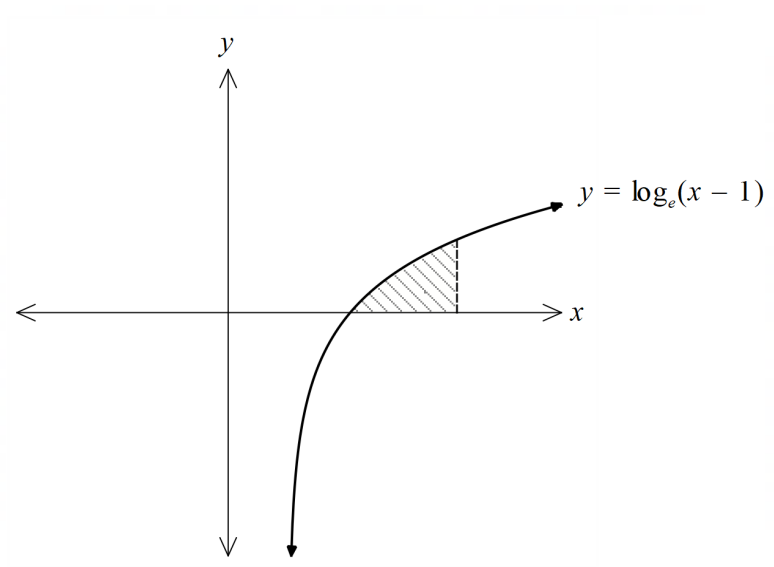
- (i) Find the coordinates where the graph crosses the x and y axes.
- (ii) State the domain of the relation.
- (iii) Find the cartesian equation of the relation.

Question 13 continues on page 9

Question 13 (continued)

Marks

- (d) Part of the graph of $y = \log_e(x-1)$ is given.



- (i) Find the value of y when $x = e+1$. **1**
- (ii) Find the shaded area, bounded by the curve, the x -axis and the line $x = e+1$. **2**

End of Question 13

Question 14 (12 marks)**Marks**

- (a) An area $A \text{ cm}^2$ is occupied by a bacteria colony. The colony has a growth rate modelled by the logistic equation

$$\frac{dA}{dt} = \frac{1}{25} A(50 - A) \text{ where } t \geq 0 \text{ and is measured in days.}$$

At time $t = 0$, the area occupied by the bacteria colony is $\frac{1}{2} \text{ cm}^2$.

(i) Show that $\frac{1}{A(50 - A)} = \frac{1}{50} \left(\frac{1}{A} + \frac{1}{50 - A} \right)$. **1**

(ii) Solve the logistic equation and hence show that $A = \frac{50}{1 + 99e^{-2t}}$. **3**

(iii) What is the limiting area of the bacteria colony? **1**

- (b) A Maths club has $2n$ members, half of which are male and half female. **2**
 A Pi Day committee is made up of 3 members of this club. By considering all different combinations for this committee, show that

$$n^n C_2 + {}^n C_3 = \frac{1}{2} {}^{2n} C_3, \text{ where } n \geq 3.$$

- (c) (i) Show that $\cos 4\theta = 8\cos^4 \theta - 8\cos^2 \theta + 1$. **2**

- (ii) By using part (i) and letting $x = \cos \theta$ in the equation $16x^4 - 16x^2 + 2 - \sqrt{2} = 0$ **3**
 show that $\cos^2 \frac{\pi}{16} \cos^2 \frac{7\pi}{16} = \frac{2 - \sqrt{2}}{16}$.

END OF PAPER



NSW Education Standards Authority

2022 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

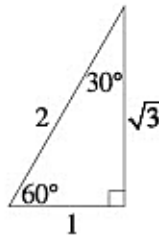
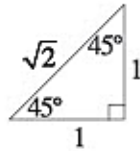
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1+t^2}$$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2} [\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2} (1 - \cos 2nx)$$

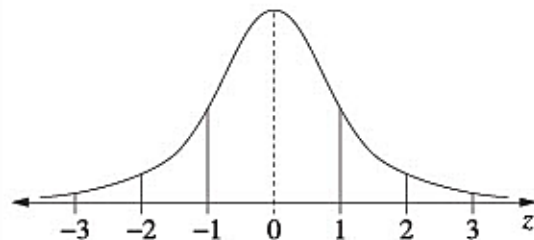
$$\cos^2 nx = \frac{1}{2} (1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq r) = \int_a^r f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1-p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= {}^nC_x p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1-p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{ f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})] \}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^n P_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x\underline{i} + y\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1 x_2 + y_1 y_2,$$

$$\text{where } \underline{u} = x_1 \underline{i} + y_1 \underline{j}$$

$$\text{and } \underline{v} = x_2 \underline{i} + y_2 \underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= r e^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n (\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2 x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$

Name:..... *Solutions*

Student
Number:

Teacher's Name:.....



Pymble Ladies' College

Mathematics Extension 1

HSC Trial Examination

Term 3 2023

ANSWER BOOKLET

- Put your name, Student Number and Teacher's name on this Booklet.
- For Questions 1-10 use the Multiple-Choice Answer Sheet provided.
- Answer Questions 11-14 in the spaces allocated for each question as these spaces provide guidance for the expected length of your response.
- Write using a non-erasable black or blue pen. Black is preferred.
- Extra writing space is provided at the end of each question.
- Calculators approved by NESA may be used.

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|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1 | A ☒ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☒ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☒ |
| 4 | A ☐ | B ☒ | C ☐ | D ☐ |
| 5 | A ☐ | B ☒ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☒ |
| 7 | A ☐ | B ☐ | C ☒ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☒ |
| 9 | A ☐ | B ☐ | C ☒ | D ☐ |
| 10 | A ☒ | B ☐ | C ☐ | D ☐ |

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SECTION II 60 marks

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is provided at the end of each question.

Question 11 (17 marks)

Marks

(a) $\frac{1}{\alpha\beta\gamma} + \frac{1}{\alpha\beta\delta} + \frac{1}{\beta\gamma\delta} + \frac{1}{\alpha\gamma\delta}$

2

$$= \frac{\delta + \gamma + \alpha + \beta}{\alpha\beta\gamma\delta}$$

$$\alpha + \beta + \gamma + \delta = 11/2$$

$$\alpha\beta\gamma\delta = 3/2$$

$$= \frac{11}{3}$$

2	correct
1	one error in process

(b) $\int \frac{1}{4x^2 + 9} dx$

2

$$= \frac{1}{4} \int \frac{1}{x^2 + 9/4} dx$$

$$= \frac{1}{4} \times \frac{1}{3/2} + \arctan \frac{x}{3/2} + C$$

$$= \frac{1}{6} + \arctan \frac{2x}{3} + C$$

2	correct
1	constant correct or $\frac{2x}{3}$ correct

Question 11 continues on page 2

(c)

2

Let the side length be x

$$\frac{dx}{dt} = 0.05 \text{ cm/s}$$

$$A = x^2$$

$$\frac{dA}{dx} = 2x$$

2	correct
1	Expression for Area, derivative and related rate correct

$$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$$

$$= 2 \times 3.1 \times 0.05$$

$$= 0.31 \text{ cm}^2/\text{s}$$

(d)

3

$$\int_0^{\pi/6} \sin 4x \cos 2x \, dx$$

$$= \frac{1}{2} \int_0^{\pi/6} (\sin(4x + 2x) + \sin(4x - 2x)) \, dx$$

$$= \frac{1}{2} \int_0^{\pi/6} (\sin 6x + \sin 2x) \, dx$$

$$= \frac{1}{2} \left[-\frac{\cos 6x}{6} - \frac{\cos 2x}{2} \right]_0^{\pi/6}$$

$$= \frac{1}{2} \left\{ -\frac{\cos \pi}{6} - \frac{\cos \pi/3}{2} - \left(-\frac{\cos 0}{6} - \frac{\cos 0}{2} \right) \right\}$$

$$= \frac{1}{2} \left\{ \frac{1}{6} - \frac{1}{4} + \frac{1}{6} + \frac{1}{2} \right\}$$

$$= \frac{7}{24}$$

3	Correct
2	Correct use of products to sums and integration error in calc
1	Collect sums to products.

Question 11 continues on page 3

(e)

(i)

1

$$|\underline{a}| = \sqrt{(-2)^2 + 3^2} \quad 1 \text{ r/w}$$

$$= \sqrt{13}$$

(ii)

1

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta = x_1 x_2 + y_1 y_2 \quad 1 \text{ r/w}$$

$$= -2 \times -6 + 3 \times 4$$

$$= 24$$

(iii)

1

$$\sqrt{13} \times \sqrt{52} \cos \theta = 24 \quad 1 \text{ r/w}$$

$$\cos \theta = 12/13$$

$$\theta = 22.61986495$$

$$\approx 23^\circ \text{ (nearest deg)}$$

(iv)

2

$$\text{Proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$$

$$= \frac{24}{52} (-6\underline{i} + 4\underline{j})$$

$$= \frac{6}{13} (-6\underline{i} + 4\underline{j})$$

2	Correct
1	Error in Calculation

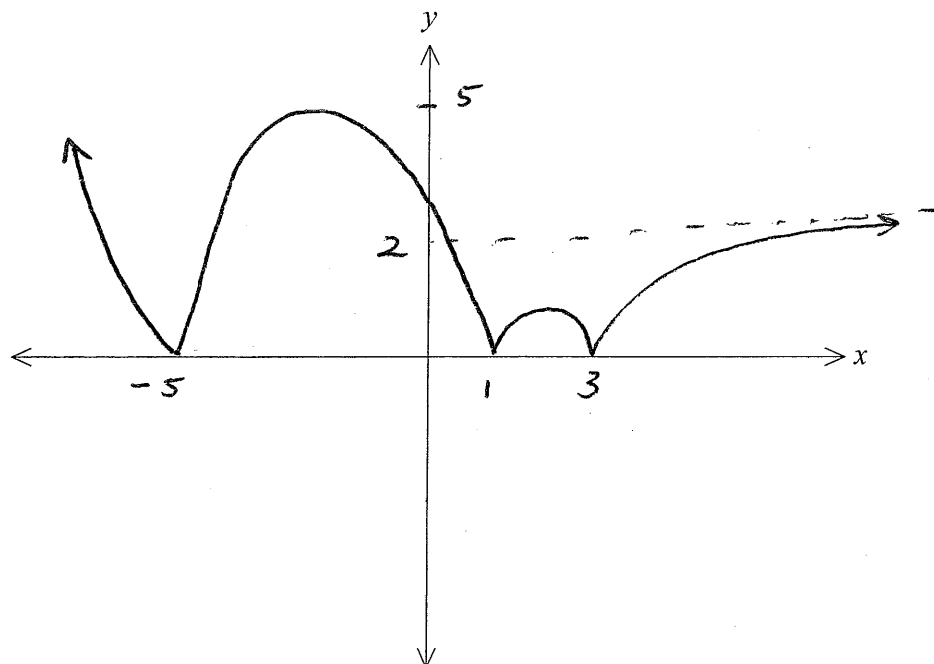
Question 11 continues on page 4

(f)

(i)

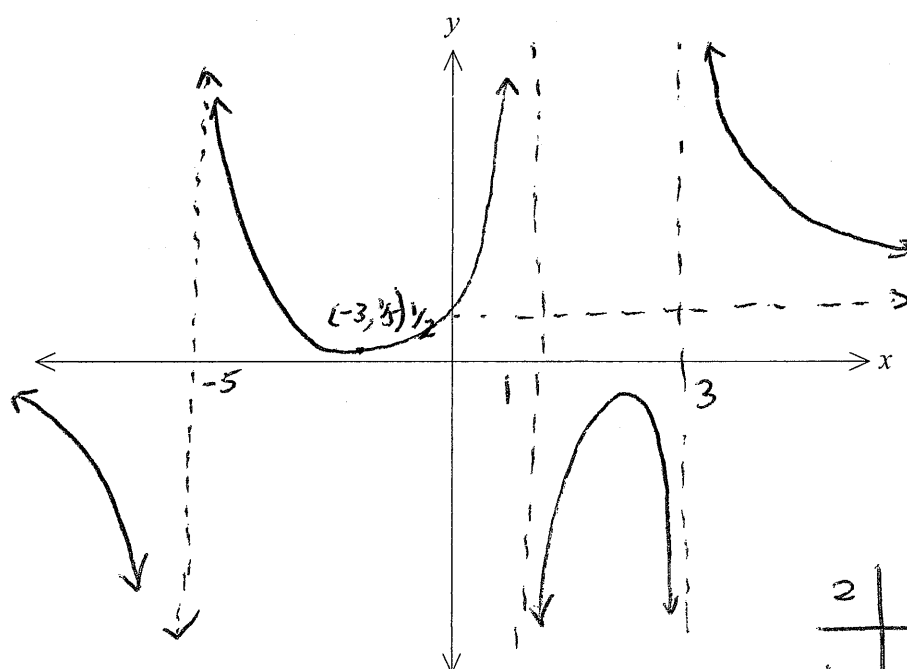
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1



(ii)

2



End of Question 11

2	correct
1	correct asymptotes and some pieces correct

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Question 12 (17 marks)

Marks

(a)

2

$$\int \frac{3x}{\sqrt{9-x^2}} dx$$

$$u = 9 - x^2$$

$$du = -2x dx$$

$$* = \frac{3}{-2} \int \frac{du}{\sqrt{u}}$$

$$= -\frac{3}{2} [2u^{1/2}] + C$$

$$= -3\sqrt{u} + C$$

$$= -3\sqrt{9-x^2} + C$$

2	Correct
1	Correct substit to *

(b)

3

$$\sqrt{3} \cos x - \sin x = A \cos(x + \alpha)$$

$$= 2(\cos x \cos \alpha - \sin x \sin \alpha)$$

$$A = \sqrt{(\sqrt{3})^2 + (-1)^2}$$

$$= 2$$

$$\therefore \sqrt{3} \cos x = 2 \cos x \cos \alpha$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$-\sin x = -2 \sin x \sin \alpha$$

$$\sin \alpha = \frac{1}{2}$$

$$\therefore \alpha = \pi/6$$

$$\therefore \sqrt{3} \cos x - \sin x = 2 \cos(x + \pi/6)$$

$$2 \cos(x + \pi/6) = -1$$

$$\cos(x + \pi/6) = -1/2$$

$$x + \pi/6 = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore x = \pi/2, \frac{7\pi}{6}$$

3	Correct
2	Correct A and α and progress towards sol
1	Correct A and α

Question 12 continues on page 8

(c)

2

(i)

$$r'(t) = (V \cos \theta) \hat{i} + (V \sin \theta - gt) \hat{j}$$

Maximum height occurs when $\hat{j} = 0$

$$\therefore V \sin \theta - gt = 0$$

$$t = \frac{V \sin \theta}{g}$$

$$\text{when } t = \frac{V \sin \theta}{g} \quad y = \frac{V \sin \theta}{g} \times V \sin \theta - \frac{1}{2} g \left(\frac{V \sin \theta}{g} \right)^2$$

$$= \frac{V^2 \sin^2 \theta}{g} - \frac{g}{2} \times \frac{V^2 \sin^2 \theta}{g^2}$$

$$= \frac{2V^2 \sin^2 \theta}{2g} - \frac{V^2 \sin^2 \theta}{2g}$$

$$= \frac{V^2 \sin^2 \theta}{2g} \quad \text{As req'd}$$

2	Correct
1	Finding t value

Question 12 continues on page 9

(c)

(ii)

2

$$\text{When } y = 0 \quad Vt \sin \theta - \frac{1}{2}gt^2 = 0$$

$$t(V \sin \theta - \frac{g}{2}t) = 0$$

$$\therefore t = 0 \quad \text{or} \quad \frac{g}{2}t = V \sin \theta$$

$$t = \frac{2V \sin \theta}{g}$$

$$\text{When } t = \frac{2V \sin \theta}{g} \quad x = V \cos \theta \times \frac{2V \sin \theta}{g}$$

$$x = \frac{V^2}{g} \sin 2\theta$$

$$\therefore \text{The range is } \frac{V^2}{g} \sin 2\theta$$

2	Correct
1	Find t

Question 12 continues on page 10

(d)

Let the statement be $4^n + 14$ is divisible by 6 for all $n \geq 1$

When $n=1$

$4^1 + 14 = 18$ which is divisible by 6

Assume the statement is true for $n=k$

i.e. $4^k + 14 = 6M$ $M \in \mathbb{Z}$, $k \geq 1$

RTP the statement is true for $n=k+1$

i.e. $4^{k+1} + 14$ is divisible by 6

Now $4^{k+1} + 14$

$$= 4 \times 4^k + 14$$

From assumption, $4^k = 6M - 14$

Substituting

$$\therefore = 4 \times (6M - 14) + 14$$

$$= 24M - 56 + 14$$

$$= 24M - 42$$

$$= 6(4M - 7)$$

Since $M \in \mathbb{Z}$, $4M - 7 \in \mathbb{Z} \therefore 4^{k+1} + 14$ is divisible by 6

Hence, since the statement is true for $n=1, k$ and $n=k+1$, the statement is true for $n=1+1=2$ and so on by the principle of mathematical induction.

3	correct
2	Set up, conclusion and some progress
1	Set up and conclusion

Question 12 continues on page 11

(e)

(i)

$$P(0) = 0 \quad \therefore d = 0$$

$$P(4) = 0 \quad \therefore 64 + 16b + 4c = 0$$

$$16b + 4c = -64$$

$$4b + c = -16 \quad \text{--- (1)}$$

$$P(-4) = 0 \quad \therefore -64 + 16b - 4c = 0$$

$$16b - 4c = 64$$

$$4b - c = 16 \quad \text{--- (2)}$$

① + ②

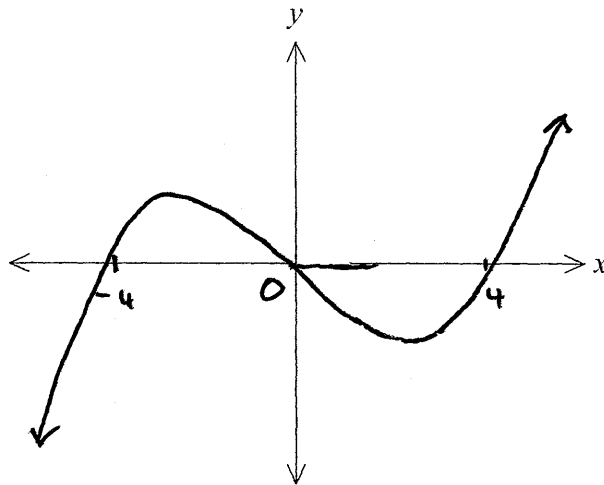
$$8b = 0 \quad \text{From (1)}$$

$$\therefore b = 0$$

$$c = -16$$

$$\therefore P(x) = x^3 - 16x$$

(ii)



$$P(x) = x(x^2 - 16) \\ = x(x-4)(x+4)$$

1 r/w

(iii)

$$\frac{x^2 - 16}{x} \leq 0$$

This is the same as finding where

$$\frac{P(x)}{x^2} \leq 0, \quad \text{since } x^2 \geq 0, \quad \text{the solution}$$

to the inequality is $x \leq -4$ or $0 < x \leq 4$

2 correct

1 Some progress

End of Question 12

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Question 13 (14 marks)

Marks

(a)
$$\left(2x + \frac{5}{x^2}\right)^6 = \sum_{r=0}^6 {}^6C_r (2x)^r (5x^{-2})^{6-r} = \sum_{r=0}^6 {}^6C_r 2^r 5^{6-r} x^{3r-12}$$

2	correct	x^3 term $\Rightarrow {}^6C_5 2^5 \cdot 5^1 x^3$
1	Working towards either x^3 or constant term	$\text{Constant term} \Rightarrow {}^6C_4 2^4 \cdot 5^2 x^0$ $\therefore \text{Term independent of } x \text{ in } \left(2 - \frac{3}{x^3}\right) \left(2x + \frac{5}{x^2}\right)^6$ $= -3 \times 6 \times 25 \times 5 + 2 \times 15 \times 2^4 \times 5^2$ $= -2880 + 12000$ $= 9120$

(b)

(i)
$$\begin{aligned} \vec{AB} &= \vec{AM} + \vec{MB} & \vec{BC} &= \vec{AC} - \vec{AB} \\ &= \vec{a} - \vec{b} & &= 2\vec{a} - (\vec{a} - \vec{b}) \end{aligned}$$

2 correct
$$= \vec{a} + \vec{b}$$

1 Just one of the vectors $\vec{AB}$ or $\vec{BC}$

(ii) Since $\triangle ABC$ is right-angled
$$|\vec{AC}|^2 = |\vec{AB}|^2 + |\vec{BC}|^2$$

$$4|\vec{a}|^2 = |\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2$$

2 correct
$$4|\vec{a}|^2 = |\vec{a}|^2 + 2|\vec{a}||\vec{b}| + |\vec{b}|^2 + |\vec{a}|^2 - 2|\vec{a}||\vec{b}| + |\vec{b}|^2$$

1 correct expansion of Pythagoras' in vector form
$$\begin{aligned} 2|\vec{a}|^2 &= 2|\vec{b}|^2 \\ |\vec{a}|^2 &= |\vec{b}|^2 \quad \text{Since } |\vec{a}| > 0 \text{ and } |\vec{b}| > 0 \\ |\vec{a}| &= |\vec{b}| \quad \text{and since } M \text{ is the midpoint of } \vec{AC} \\ \vec{AM} &= \vec{MC} = \vec{a} \end{aligned}$$

$\therefore A, B, C$ are equidistant from M

Question 13 continues on page 16

(c)

2

(i) $x = 1 - \sin t$ $y = \cos t \sin 2t$

When $x = 0$ When $y = 0$

2 correct

1 Letting
 $x, y = 0$
and
solving
for t

$$\sin t = 1$$

$$t = \pi/2$$

$$\cos t = 0 \text{ or } \sin 2t = 0$$

$$t = \pi/2 \text{ or } 2t = 0, \pi, 2\pi$$

$$t = 0, \pi/2, \pi$$

When $t = 0$ $x = 1$ $y = 0$ $\therefore x$ intercepts

$t = \pi/2$ $x = 0$ $y = 0$ are 1 and 0.

$t = \pi$ $x = 1$ $y = 0$ y intercept
is 0

(ii)

1

$$x = 1 - \sin t \quad 0 \leq t \leq \pi$$

Domain

$$0 \leq x \leq 1$$

1 r/w

(iii)

2

2 correct

1 Progress

towards

removing

t

$$y = \cos t \times 2 \sin t \cos t$$

$$= 2 \sin t \cos^2 t$$

$$\sin t = 1 - x$$

$$= 2 \sin t (1 - \sin^2 t)$$

$$= 2(1-x)(1-(1-x)^2)$$

$$= 2(1-x)(1-(1-2x+x^2))$$

$$= 2(1-x)(2x-x^2)$$

$$y = 2x(1-x)(2-x); \quad 0 \leq x \leq 1$$

$$\text{OR } y = 2x^3 - 6x^2 + 4x; \quad 0 \leq x \leq 1.$$

(d)

(i)

1

$$y = \ln(e+1-1) \quad 1 \text{ r/w}$$

$$= 1$$

$$(ii) \int_1^{e+1} \ln(x-1) dx \quad y = \ln(x-1) \quad 2$$

$$x-1 = e^y$$

$$x = e^y + 1$$

$$= (e+1) \times 1 - \int_0^1 (e^y + 1) dy$$

$$= e+1 - [e^y + y]_0^1$$

$$= e+1 - (e+1 - e^0)$$

$$= 1 \text{ unit}^2$$

or

$$u = \ln(x-1) \quad v' = 1$$

$$u' = \frac{1}{x-1} \quad v = x$$

$$A = [x \ln(x-1)]_2^{e+1} - \int_2^{e+1} \frac{x}{x-1} dx$$

$$= e+1 - 0 - \int_2^{e+1} \left(\frac{x-1}{x-1} + \frac{1}{x-1} \right) dx$$

End of Question 13

$$= e+1 - [x + \ln(x-1)]_2^{e+1}$$

$$= e+1 - [e+1+1 - (2+0)]$$

$$= 1 \text{ unit}^2$$

2	Correct
1	Some progress by finding the difference of areas or using by parts

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Question 14 (12 marks)

Marks

(a)

1

(i)

$$\text{RHS: } \frac{1}{50A} + \frac{1}{50(50-A)} \quad \text{1 r/w}$$

$$= \frac{50-A+A}{50A(50-A)}$$

$$= \frac{50}{50A(50-A)}$$

$$= \frac{1}{A(50-A)}$$

$$= \text{LHS}$$

Question 14 continues on page 22

Question 14
Part (a) continued

Marks

3

(ii) $\frac{dA}{dt} = \frac{A}{25} (50 - A)$

$$\int \frac{dA}{A(50-A)} = \int \frac{1}{25} dt$$

$$\frac{25}{50} \int \left(\frac{1}{A} + \frac{1}{50-A} \right) dA = t + C$$

$$\frac{1}{2} [\ln|A| - \ln|50-A|] = t + C$$

When $t = 0$ $A = 0.5$

$$\frac{1}{2} \left\{ \ln \left| \frac{0.5}{49.5} \right| \right\} = C$$

$$C = \frac{1}{2} \ln \frac{1}{99}$$

$$\therefore \frac{1}{2} \ln \left| \frac{A}{50-A} \right| = t + \frac{1}{2} \ln \frac{1}{99}$$

$$2t = \left(\ln \left| \frac{A}{50-A} \right| - \ln \frac{1}{99} \right)$$

$$2t = \ln \left| \frac{99A}{50-A} \right|$$

$$** e^{2t} = \frac{99A}{50-A}$$

$$e^{2t}(50-A) = 99A$$

$$50e^{2t} - Ae^{2t} = 99A$$

$$A(99 + e^{2t}) = 50e^{2t}$$

$$A = \frac{50e^{2t}}{99 + e^{2t}} \div \frac{e^{2t}}{e^{2t}}$$

$$A = \frac{50}{99e^{-2t} + 1}$$

As req'd

1 subst from
(i)

2 $e^{2t} = \frac{99A}{50-A}$
or equivalent

3 correct soln

Question 14 continues on page 23

Question 14

Marks

Part (a) continued

(iii)

1

$$As t \rightarrow \infty \quad A \rightarrow \frac{50}{0+1} \quad 1r/w$$

$$\rightarrow 50 \text{ cm}^2$$

(b)

2

$n \rightarrow \text{female} \quad n \rightarrow \text{male}$
 Pi Day Committee: 3 male, 2 male 1 female,
 1 male 2 female, 3 female

$$\therefore {}^nC_3 + {}^nC_2 \times {}^nC_1 + {}^nC_1 \times {}^nC_2 + {}^nC_3$$

$$= 2 \times {}^nC_3 + 2 \times {}^nC_2 \times {}^nC_1$$

$${}^nC_1 = n$$

$$\therefore = 2({}^nC_3 + n \times {}^nC_2)$$

$$\text{Pi Day Committee: } 2{}^nC_3$$

$$2{}^nC_3 = 2({}^nC_3 + n \times {}^nC_2)$$

$$\frac{1}{2} 2{}^nC_3 = {}^nC_3 + n \times {}^nC_2 \quad \text{As eq'd}$$

Lists out-comes
 3M nC_3
 2M1F ${}^nC_2 \times {}^nC_1$
 1M2F ${}^nC_1 \times {}^nC_2$
 3M nC_3

2 correct soln

Question 14 continues on page 24

(c)

2

(i)

$$\begin{aligned}
 \cos 4\theta &= \cos(2\theta + 2\theta) \\
 &= \cos^2 2\theta - \sin^2 2\theta \\
 &= \cos^2 2\theta - (1 - \cos^2 2\theta) \\
 &= 2\cos^2 2\theta - 1 \\
 &= 2(\cos^2 \theta - \sin^2 \theta) - 1 \\
 &= 2(\cos^4 \theta - 2\cos^2 \theta \sin^2 \theta + \sin^4 \theta) - 1 \\
 &= 2(\cos^4 \theta - 2\cos^2 \theta(1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2) - 1 \\
 &= 2(\cos^4 \theta - 2\cos^2 \theta + 2\cos^4 \theta + 1 - 2\cos^2 \theta + \cos^4 \theta) - 1 \\
 &= 2(4\cos^4 \theta - 4\cos^2 \theta + 1) - 1
 \end{aligned}$$

1 applies correct identity

2 correct soln

$$\begin{aligned}
 \cos 4\theta &= 8\cos^4 \theta - 8\cos^2 \theta + 2 - 1 \\
 \cos 4\theta &= 8\cos^4 \theta - 8\cos^2 \theta + 1 \quad \text{As req'd}
 \end{aligned}$$

(ii)

3

$$\text{Let } x = \cos \theta$$

$$16\cos^4 \theta - 16\cos^2 \theta + 2 - \sqrt{2} = 0$$

$$8\cos^4 \theta - 8\cos^2 \theta + 1 = \frac{\sqrt{2}}{2}$$

* $\therefore \cos 4\theta = \frac{\sqrt{2}}{2}$ $\therefore$ solving the eqⁿ is equivalent to finding the 4 distinct values of $\cos 4\theta = \frac{\sqrt{2}}{2}$

$$4\theta = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}$$

$$\theta = \frac{\pi}{16}, \frac{7\pi}{16}, \frac{9\pi}{16}, \frac{15\pi}{16}$$

$$\cos \frac{9\pi}{16} = -\cos \frac{7\pi}{16} \quad \cos \frac{15\pi}{16} = -\cos \frac{\pi}{16}$$

1 substitute $\cos \theta$ and manipulate eqn

$$\text{Roots are } \pm \cos \frac{\pi}{16}, \pm \cos \frac{7\pi}{16}$$

$$\text{Product of roots } \cos \frac{2\pi}{16} \cos \frac{7\pi}{16} = \frac{2 - \sqrt{2}}{16}$$

$$2 \quad \theta = \frac{\pi}{16}, \frac{7\pi}{16}, \frac{9\pi}{16}, \frac{15\pi}{16}$$

must have 4 solns

As req'd

3 correctly reaches conclusion

If you use this space, clearly indicate which question you are answering.

4 c) i) quicker solution

$$\begin{aligned}\cos 4\theta &= 2\cos^2 2\theta - 1 \\ &= 2[2\cos^2 \theta - 1]^2 - 1 \\ &= 2(4\cos^4 \theta - 4\cos^2 \theta - 1) - 1 \\ &= 8\cos^4 \theta - 8\cos^2 \theta - 2 + 1 \\ &= 8\cos^4 \theta - 8\cos^2 \theta - 1 \\ &= \text{RHS as required}\end{aligned}$$

